

THOMAS AMOGOLLA TSUMA

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Education

MULTIMEDIA UNIVERSITY OF KENYA

Bachelor of Science, Computer Science

Second Class Upper Honors

Projects: Lung Cancer Detection in CT Scans, Hate Speech Detection using NLP

Relevant Coursework: Learning and Adaptive Systems, Artificial Intelligence, Software Engineering

INTI INTERNATIONAL UNIVERSITY

Master of Research, Data Science

Research: Generative Adversarial Networks, Graph Neural Networks

Experience

CROP NUTRITION LABORATORY SERVICES

Limuru, Kenya

Machine Learning Engineer

September 2022

- Developed an end-to-end automation pipeline for artificial neural network (ANN) optimization, sourcing soil spectra from PostgreSQL and wet-chemical results from MySQL, cleaning data, and versioning it with DVC before uploading to S3 and deploying DigitalOcean droplets via Bash. This system standardized model updates with a 4-day turnaround time.
- Collaborated on building a comprehensive evaluation tool for soil spectroscopy models, integrating scatterplot visualizations and advanced metrics including RMSE, R², RSC, slope, intercept, kurtosis, and skewness to analyze model performance.
- Designed and trained a spectral outlier detection system using an RNN Autoencoder, reducing prediction delays by automatically filtering erroneous spectra before inference.
- Volunteered mobile development support during downsizing, addressing critical bugs and implementing new features to maintain application stability in the absence of the dedicated mobile team.
- Built a multi-label Random Forest classifier to identify plant types and growth stages, helping lab technicians validate sample accuracy prior to testing and minimizing misclassification errors.
- Deployed trained models as Azure Functions on Microsoft Azure ML, enabling seamless integration and access for internal developers.
- Designed and developed deep learning models using the concept of pix2pix generative adversarial networks to generate synthetic alpha-atr spectra. This helped land the company a contract with KALRO to analyze over 60,000 spectra which resulted in an influx of 3M USD in revenue
- Developed an EPO conversion workflow to allow 2mm alpha-atr spectra to mimic fritsch-milled alpha-atr spectra. This allowed us to analyze samples from Loyola University without rescanning the spectra

PAYLEND LIMITED

Nairobi, Kenya

MOBILE DEVELOPMENT | AI ENGINEER

January 2022 – September 2022

- Created and launched the second version of the Paylend mobile application, integrating expanded functionalities including business loans, school fees loans, personal loans, fundraising, and utility bill payments extending our reach to three countries.
- Engineered an object identification model using TensorFlow to detect identity cards in user-uploaded images and match facial data with profile pictures, enhancing the platform's layered authentication system which resulted in the early flagging of fraudulent accounts and reduced onboarding time for clients to a few hours from nearly two days.
- Formulated and implemented an RFM (Recency, Frequency, Monetary) analysis model to quantify customer behaviors and determine data-driven loan limits, replacing arbitrary credit decisions.
- Developed a seasonal trend forecasting model using ARIMA to analyze the temporal patterns of loan issuance, directly correlating with revenue and customer growth; this insight informed strategic advertising cycles.
- Built a credit risk model using logistic regression to predict potential loan defaulters, which was integrated into the loan approval pipeline and reduced default rates which identified around 60% of users who defaulted after having their loan limits increased.
- Created an interactive business intelligence dashboard using Python's Dash and Plotly, and hosted it on an AWS EC2 instance for internal decision-making.
- Deployed model inference APIs using Flask and configured production-ready services with Gunicorn on Amazon EC2, streamlining access to ML models for the broader application ecosystem.

CIRCULARITY LTD

Ongata Rongai, Kenya

Machine Learning Engineer

November 2020 – July 2022

- Designed, developed, and integrated a plastic bottle identification system by combining an image classification Convolutional Neural Network (CNN) with an object detection pipeline using the TensorFlow Object Detection API.
- Adapted InceptionV3 as the pre-trained base for the image classification model, enhancing initial accuracy and reducing training time.
- Combined classification and detection models to fortify performance by mitigating individual weaknesses, resulting in a more reliable waste sorting tool.
- Produced a Minimum Viable Product (MVP) that influenced strategic partnerships, leading to a signed Memorandum of Understanding (MOU) with Coca-Cola.
- Publicized the innovation through a feature on KTN's *The Innovator*, promoting the company's visibility and securing media traction.

PROJECTS

Core&Outline – End-to-End Data Automation & Intelligence Platform

HTML, CSS, Flask, ExpressJS, SageMaker, S3, AWS Glue, AWS Lambda, EventBridge, Snowflake, PostgreSQL, Reinforcement Learning, GreatExpectations, Kafka, Flink, Spark, Text Classification, LLMs

- Designed and built a low-code analytics and automation platform to help businesses turn raw data into actionable dashboards and machine learning insights—fully automated from ingestion to visualization.

- Engineered data pipelines using AWS Lambda (deployed via EC2 containers) to pull data from sources including Snowflake, PostgreSQL, MySQL, and CSVs, temporarily storing in Parquet format on Amazon S3.
- Utilized AWS Glue and implemented GreatExpectations for automated data quality checks and transformations, populating a Snowflake warehouse with structured business metrics (e.g., MRR, churn, CTR, session rates).
- Developed real-time streaming workflows using Kafka, Flink, and Spark to track and analyze user sessions across websites and mobile apps.
- Built a reinforcement learning–based recommendation engine using click and purchase session data to drive personalized suggestions directly from the platform’s dashboard.
- Created a text classification model to pre-route customer complaints (e.g., account vs. shipping issues) into LLM-powered conversational AI agents, hosted at [Hugging Face](#).
- Authored technical content on the platform’s blog, showcasing use cases in data engineering, ML, and automation: [Core&Outline Blog](#).

Rydr – Route-Based Ride Matching App

React Native, Node.js, Express.js, IBM Cloud, Google Maps API

- Developed and launched a mobile application on the Google Play Store that matches passengers and drivers based on route similarity and departure times.
- Built and deployed a backend API using Express.js on IBM Cloud Foundry, enabling real-time ride coordination powered by the Google Maps API.

PUBLICATIONS

AI Agents for Supercharged Supply Chain and Inventory Management

Reinforcement Learning | TensorFlow | Linear Algebra

[Read Article](#)

Authored and published on the Core&Outline blog. The article explores the use of Deep Q-Learning with Shaped Rewards to optimize inventory management. It’s structured in four layers—business objectives, mathematical modeling, code implementation, and real-world impact—to engage both technical and non-technical stakeholders.

Lung Cancer Diagnosis: Making Machines That Think Like Human Beings

Convolutional Neural Networks | TensorFlow | Keras | Transfer Learning | OpenCV

Read Article

Published by Analytics Vidhya. Demonstrated how deep learning models can analyze lung CT scans to detect cancerous nodules. Combined data acquisition, cleaning, and augmentation with CNNs and transfer learning for medical image classification. Included extensions like time series forecasting, anomaly detection, and knowledge graphs for advanced medical analytics.

Coca-Cola Bottle Image Recognition

Image Segmentation | CNNs | TensorFlow | Keras | OpenCV

[Read Article](#)

Published in the Analytics Vidhya Data Science Blogathon, ranked Top 30 out of 400+ submissions globally. Demonstrated how to identify branded bottles using color segmentation, transfer learning, and image classification techniques in TensorFlow and OpenCV.

Leadership and Activities

ALLIANCE HIGH SCHOOL KNIGHTS BASKETBALL

Kiambu, Kenya

Basketball Captain

2016

- Led my team to its first national finals appearance in over two decades and earned the school a chance to represent Kenya in the East African Games as well.

Awards and Certifications

AWS | AWS Machine Learning Specialty (ML-CS01)

MICROSOFT | Azure AI Fundamentals (ai-900)

IBM | Data Science Fundamentals

HUAWEI | Huawei Certified Networking Associate

Technical Skills & Abilities

Languages: Python, Java, Dart, JavaScript, SQL, C, C++, R, JSON

Frameworks: Tensorflow, Keras, Pytorch, NLTK, Sci-kit Learn, Flask, Django, Android, Flutter, React, React Native, NodeJS, ExpressJS, MySQL, PostgreSQL,

Cloud and Security Tools: AWS Cloud, DigitalOcean, IBM Cloud, Azure Cloud, SQL Server, Google Cloud

Platform

DevOps and API Tools: Git, Docker, AWS CodeDeploy (CI/CD), Swagger,